

INNOVATION, IMPLEMENTATION, IMPROVEMENT OPEN ACCESS

Implementing an Escape Room for Clinical Skills Practice in a Paediatrics Clerkship

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ABSTRACT

Background: Escape rooms include the same experiential learning methods that make traditional simulation practices effective while adding gamification which improves learner engagement. Our existing simulation curriculum lacked active experimentation, an essential component of Kolb's experiential learning cycle. The goal of this curriculum innovation was to determine the feasibility of adding an escape room immediately following the simulation to provide an effective independent learning experience for learners.

Approach: We created a 40-min escape room activity using principles of self-determination theory. All students in the paediatric clerkship participated once over one academic year. Students were presented with an unstable child with seizures and worked in teams of 3–5 to open locks and solve puzzles as they applied nine clinical skills learned from the preceding simulation session.

Evaluation: Students ($N=66$) reported increases in knowledge after both the simulation and the escape room and an increase in confidence after completing the escape room, indicating we were successful in adding active experimentation to our curriculum. They also commented on the value of applying and practising new skills and of engaging in teamwork, which reinforced the utilization of self-determination theory as a guiding principle for escape room design.

Implications: Our escape room is novel in that it takes place in a paediatric clerkship setting and builds upon existing curricular elements to provide students with independent practice of clinical skills in a gamified and fun approach early in their clinical training years. Escape rooms are a feasible and an effective method to provide hands-on experience to clinical learning.

1 | Background

Simulation has a long history in medical education and continues to evolve as new technologies and approaches become available [1]. Escape rooms, originally created as a leisure activity, are a recent addition to healthcare simulation. They require participants to work in teams in a simulated environment to solve clinically based puzzles under time pressure to 'escape' the room [2, 3]. Escape rooms harness the same experiential learning methods that make traditional simulation practices effective while adding gamification to increase

learner engagement, improve problem solving, and foster goal achievement [4–6].

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Our institution used facilitator-assisted simulation to teach foundational clinical skills to medical doctorate students during the

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paediatrics clerkship. These sessions were intended to introduce the learners to new skills and provide limited, assisted hands-on practice. The facilitator-assisted simulation included the first three parts of Kolb's experiential learning cycle (Figure 1), providing students with opportunities for concrete experiences, reflective observation, and abstract conceptualization [7,8]. However, we realised the fourth part of Kolb's cycle, active experimentation, was lacking from our curriculum because learners were supported by a facilitator during the simulation and did not have the independent practice that is a necessary part of experiential learning.

To address this gap, we added an escape room activity following the existing simulation to allow students independent clinical skill practice. We chose an escape room as opposed to other experiential learning modalities because it was a novel approach for paediatric medical students that was backed by an established theoretical underpinning, namely self-determination theory, which recognises that adults are motivated when their needs for competence, relatedness and autonomy are met [9]. Additionally, previous studies indicate that escape rooms are effective for teaching teamwork and communication, two essential skills for medical learners [10]. This educational case study describes the implementation and evaluation of the escape room in our paediatric clerkship. We used pre-simulation, post-simulation, and post-escape room knowledge and confidence surveys regarding the clinical learning objectives, as well as open-ended survey questions, to determine the feasibility of an escape room in providing an effective active experimentation experience for learners.

2 | Approach

Our escape room took place during the 2022–2023 academic year and included 80 medical doctorate students on their 8-week paediatric clerkship rotation. The escape room activity repeated once a week for each of the eight weeks of the clerkship, and all

learners participated once, in groups of three to five. The activity is built around the clinical scenario of a toddler who presents with seizures and is found to be unstable. Throughout the activity, students are prompted to perform nine specific skills that were introduced during the preceding simulation activity (Figure 2). The activity was deemed not human subjects research by the Institutional Review Board.

We use the medical school's simulation lab for the escape room activity, which is equipped with manikins, mounted screens, video feed, speakers, and microphones, all of which can be remotely operated. A simulation technician operates the manikin that plays the patient and receives specific instructions on how to adjust its vitals and movements as the students progress through the escape room case.

The session begins with a 5 min orientation to the escape room led by a faculty member. The orientation explains that learners will be working together to solve a series of puzzles and that teamwork skills are necessary to be successful. This helps establish **relatedness** as a key component of the activity. Afterwards, the students are left on their own to begin the activity.

The scenario unfolds over 40 min, with students correctly performing clinical tasks in order to unlock puzzle boxes which provide them with new information required to proceed with the activity. The average completion time per group was 22 min and 34s; the maximum time was 35 min and 5s whereas the minimum time was 14 min and 49s. Faculty facilitators watched the learners remotely and were able to provide hints via microphones placed in the room if they became stuck. These tasks foster **competence** in the learners, as they can demonstrate skills they previously learned without direct faculty assistance. They also provide opportunities for **autonomy** since students choose what role they will play and bring their own unique strengths to each puzzle.

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After the students have completed all the tasks and escaped the room, the facilitator re-enters the room and asks the students to provide a patient assessment. The facilitator then conducts a 7–10 min debrief session with the learners, asking students to reflect on areas where they may have gotten stuck, in addition to discussing their differential diagnosis, work up and management.

3 | Evaluation

Students completed surveys at three different stages during the simulation and escape room activities: pre-simulation, post-simulation, and post-escape room. These surveys assessed students' knowledge, skills, and confidence regarding the educational objectives of the activities. The methodology included:

- Knowledge Assessment: Students responded to short-answer questions, which were later graded by the authors.

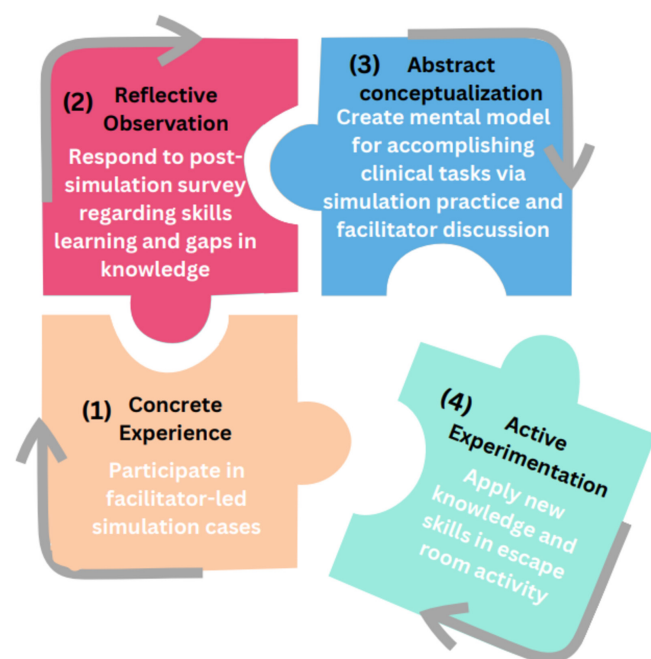


FIGURE 1 | Application of Kolb's experiential learning cycle.

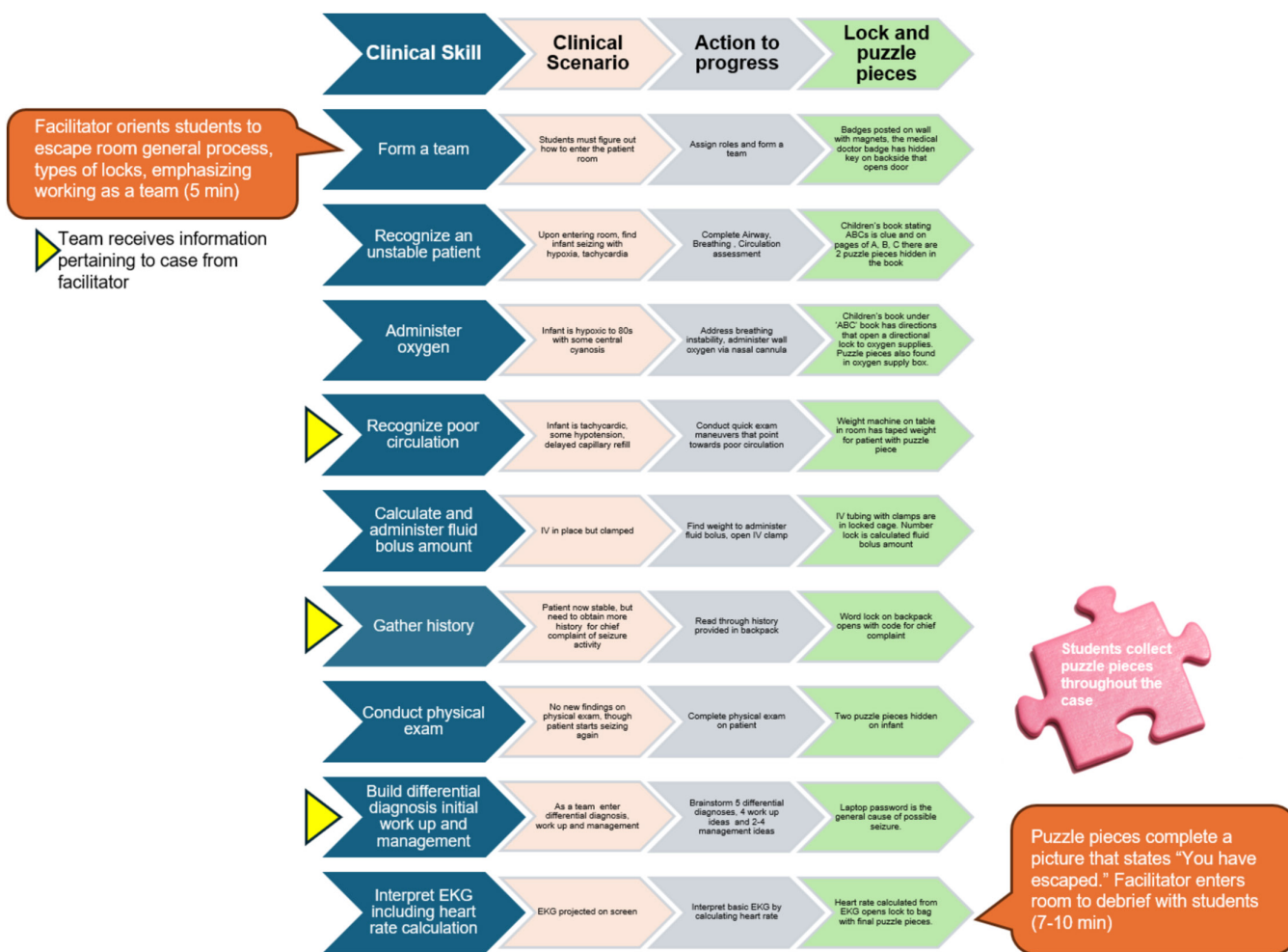


FIGURE 2 | Flow diagram of escape room clinical tasks.

Responses were converted to non-parametric categories: completely correct, partially correct, or incorrect. This allowed us to assess knowledge retention at Kirkpatrick Level 2 [11].

- **Skills and Confidence Evaluation:** A Likert scale (ranging from 1, indicating 'not confident' to 4, representing 'most confident') was employed to gauge students' self-assessed skills and confidence levels. These questions were focused on the skills learned during the simulation and applied in the escape room. This allowed us to assess for confidence in skills at Kirkpatrick Level 1 [11].
- **Student Feedback:** To understand the most effective aspects of the escape room and identify areas for curricular improvement, we collected comments from open-ended survey questions.

Statistical analyses were conducted using R and Python software. The data were analysed using ANOVA for parametric measures and Friedman's analysis for non-parametric measures. Additionally, pairwise comparisons of the pre-simulation, post-simulation, and post-escape room results were conducted using the Durbin-Conover test.

Eighty students participated in the simulation and escape room; 77, 73, and 66 students respectively completed the

pre-simulation, post-simulation, and post-escape room surveys. Students reported increases in knowledge after the simulation session and retained that knowledge after the escape room. While students generally reported increases in confidence after the simulation session, their confidence in applying their skills was further increased after the escape room activity (Table 1). In open-ended survey responses, students indicated that they found the application and practice of new skills, fun, teamwork, game-like experience, and the fast pace to be among the most valuable aspects of the escape room (Table 2).

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Taken together, this survey data demonstrate that implementing an escape room is an effective way to add the final Kolb's cycle step of active experimentation to our curriculum and closes the curricular gap we identified. They indicate that learners not only gained confidence in their new skills, but they recognised the importance of having an activity that allowed them to apply those skills while working as a team. By placing the escape room activity after the facilitated simulation, the new knowledge and skills were reinforced with independent practice; this approach aligns with previous studies that indicate escape rooms are an

TABLE 1 | Changes in learner knowledge and confidence before and after simulation and escape room activities.

Survey distributed/ completed response rate	PreSIM 80/77 96%		PostSIM 80/73 91%		PostESCAPE 80/66 83%	
	Knowledge score [†]	Mean confidence score* (SD)	Knowledge score [†]	Mean confidence Score* (SD)	Knowledge score [†]	Mean confidence score* (SD)
Clinical skills						
Initial patient assessment: Airway breathing circulation	53.2%	2.15 (0.64)	100% ^α	3.34 ^α (0.51)	100% ^β	3.44 ^β (0.56)
Oxygen delivery devices	11.7%	1.82 (0.49)	69.9% ^α	2.98 ^α (0.57)	71.2% ^β	3.28 ^β (0.66)
Assessing circulation/ recognising signs of dehydration	10.4%	2.28 (0.58)	84.9% ^α	3.23 ^α (0.54)	83.3% ^β	3.43 ^{β,γ} (0.56)
EKG heart rate measurement	36.4%	1.77 (0.65)	68.5% ^α	2.61 ^α (0.65)	65.2% ^β	3.03 ^{β,γ} (0.64)
Fluid resuscitation calculation	40.3%	1.69 (0.74)	98.6% ^α	3.15 ^α (0.67)	98.5% ^β	3.36 ^β (0.63)

[†]Non-parametric measures (Friedman test), score reflects percentage students who got completely correct.

*Confidence scores on Likert scale 1–4; 1 = not at all confident and 4 = most confident.

^αp-value < 0.05 comparing pre-simulation scores to post-simulation scores.

^βp-value < 0.05 comparing pre-simulation scores to post escape room scores.

^γp-value < 0.05 comparing post-simulation to post escape room scores.

TABLE 2 | Top student responses to question ‘What was most valuable about this ESCAPE room style SIM?’

Response category	% of responses	Example quotation
Application and practice of new skills	35%	<i>Applying what we learned, was a fun way to reason through it without as much attending help</i>
Fun	31%	<i>It was a fun way to incorporate the knowledge we learned</i>
Teamwork	17%	<i>Working as a team</i>
Game-like experience	9%	<i>Incorporating medical knowledge to solve clues</i>
Fast pace	7%	<i>Figuring out how to do these calculations and tasks under pressure and working as a team</i>

effective technique for applying previously taught information [12].

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Our open-ended survey responses also demonstrate that learners experienced high levels of competence and relatedness through practical application of skills and teamwork, reinforcing self-determination as the theoretical underpinning for the activity. Lastly, the learners enjoyed the activity, with fun being the number two category in the open-ended responses. Student enjoyment is a key feature of escape rooms, as demonstrated in a systematic review of healthcare escape rooms [3]. Thus, while we cannot say that the escape room is superior to other experiential learning methods, we are confident that it is a feasible and effective method for learners to apply foundational clinical skills.

“It is a feasible and effective method for learners to apply foundational clinical skills.”

4 | Implications

The escape room is now established in our paediatrics clerkship as a standard part of the curriculum. Given the wide applicability of the tasks built into the escape room, it could be modified for more advanced learners, such as paediatric residents, by adding complexity to the case that requires more complex clinical reasoning and skills. The content is also relevant for learners in ICU settings and could be incorporated into a critical care rotation for medical students or residents. The next steps in developing this innovation are to assess student knowledge and confidence retention and possibly have students engage in retrieval practice of knowledge and skills by conducting an escape room at the end of medical school as preparation for entering residency [13].

Our escape room has some limitations. Because our goal was to examine the feasibility of escape rooms for experiential learning and not to measure whether it was more advantageous than other methods, we did not use a comparison group and are therefore unable to determine if an escape room is educationally superior when compared to other experiential learning methodologies. Learners come into the clerkship with variable levels of clinical experience, which may impact their confidence and skill regarding the escape room tasks. Lastly, our study takes place at a single institution across one academic year.

Our escape room is novel because it is the first study involving a paediatric clerkship student audience that takes place in a simulation center [3, 6, 14–18]. Our escape room is also novel because it explicitly links the activity to two educational theories, experiential learning and self-determination, that are widely used in medical education, providing additional justification for escape rooms as an effective teaching method.

Author Contributions

Sofia Khara: writing – original draft, investigation, conceptualization, writing – review and editing, methodology, data curation, supervision, visualization. **Andrew Wai:** conceptualization, investigation, methodology, visualization, writing – review and editing, project administration, resources, data curation, supervision. **Christine Ford:** formal analysis, writing – original draft, writing – review and editing, investigation, visualization, data curation, methodology. **Andrew Yoo:** conceptualization, methodology, visualization, writing – review and editing, project administration, supervision, resources, data curation, investigation. **Daniel Joe:** writing – review and editing, methodology, visualization, supervision, resources, data curation, investigation. **Udochukwu Oyoyo:** investigation, writing – review and editing, formal analysis, data curation, methodology.

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Ethics Statement

Loma Linda University's Institutional Review Board concluded this educational case study did not meet the definition of human subjects research. All students entered survey responses anonymously using coded identification numbers to pair results.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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